



CY104 – Introduction to Cybersecurity

LAB # 02

Linux Commands (Basic)

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Task 01

Creating a new directory

Using mkdir, we can create a new folder:

```
(atharimran@2604277)-[~]  
$ mkdir secret  
  
(atharimran@2604277)-[~]  
$ ls  
Desktop    Downloads  Pictures   Public     Templates  
Documents  Music      Projects   secret     Videos
```

Moving inside a directory

Using cd, we can Change Directory:

```
(atharimran@2604277)-[~]  
$ cd secret/  
  
(atharimran@2604277)-[~/secret]  
$ pwd  
/home/atharimran/secret
```

Creating files and writing content

With nano command, we can create and write content into a file:

```
(atharimran@2604277)-[~/secret]  
$ nano clue1.txt
```

Ls means List, which can verify our newly created file:

```
(atharimran@2604277)-[~/secret]  
$ ls -l  
total 8  
-rw-rw-r-- 1 atharimran atharimran 73 Sep 17 12:12 clue1.txt
```

This is what I wrote in clue1.txt:

```
(atharimran@2604277)-[~/secret]
$ cat clue1.txt
This is a SECRET.
This LINE has a secret.
And this is a normal sentence.
```

Now, the same process with second file:

```
(atharimran@2604277)-[~/secret]
$ nano clue2.txt
```

```
(atharimran@2604277)-[~/secret]
$ ls -l
total 8
-rw-rw-r-- 1 atharimran atharimran 73 Sep 17 12:12 clue1.txt
-rw-rw-r-- 1 atharimran atharimran 86 Sep 17 12:13 clue2.txt
```

```
(atharimran@2604277)-[~/secret]
$ cat clue2.txt
Look for the WORD in it.
This sentence has FIVE words.
This is an abnormal sentence.
```

Task 02

In this task, we will test the backup.

Creating Project

Using mkdir, I have created a project folder:

```
(atharimran@2604277)-[~]  
$ mkdir Project
```

In this folder, I have created three files:

```
(atharimran@2604277)-[~/Project]  
$ echo "This is file 1." > file1.txt  
  
(atharimran@2604277)-[~/Project]  
$ echo "This is file 2." > file2.txt  
  
(atharimran@2604277)-[~/Project]  
$ echo "This is file 3." > file3.txt
```

Copying Project files into Backup

```
(atharimran@2604277)-[~]  
$ cp -r Project Backup  
  
(atharimran@2604277)-[~]  
$ ls -l Backup/  
total 12  
-rw-rw-r-- 1 atharimran atharimran 16 Sep 17 12:15 file1.txt  
-rw-rw-r-- 1 atharimran atharimran 16 Sep 17 12:15 file2.txt  
-rw-rw-r-- 1 atharimran atharimran 16 Sep 17 12:15 file3.txt
```

Using cp (copy), we can copy files from one folder to another. It automatically creates the other folder if it doesn't exist.

Deleting a file:

```
(atharimran@2604277)-[~]  
$ rm Project/file2.txt  
  
(atharimran@2604277)-[~]  
$ ls -l Project  
total 8  
-rw-rw-r-- 1 atharimran atharimran 16 Sep 17 12:14 file1.txt  
-rw-rw-r-- 1 atharimran atharimran 16 Sep 17 12:15 file3.txt
```

Differentiating two folders

With the diff command, we can differentiate between two folders:

```
(atharimran@2604277)-[~]  
$ diff -r Project Backup/  
Only in Backup/: file2.txt
```

It shows that file2.txt, which we deleted from the Project folder, now exists only in Backup.

Hence, differentiating between the two folders.